

Ph.D. Exam: Numerical Analysis, August, 2025
 Do 4 (four) of the first 5 (1-5) and 4 (four) of the
 last 5 problems (6-10).

1. Suppose A is Hermitian positive definite.
 - (a) Prove that each principal submatrix of A is Hermitian positive definite.
 - (b) Prove that an element of A with largest magnitude lies on the diagonal.
 - (c) Prove that A has a Cholesky decomposition.
2. Assume $A \in C^{m \times m}$.
 - (a) Show that A has a Schur decomposition.
 - (b) If A has a collection of m linearly independent eigenvectors, show that A is diagonalizable.
3. Let $A \in C^{m \times n}$, with $m \geq n$ and $\text{rank}(A) = p = n \geq 3$. Let $a_1, a_2, \dots$ denote the columns of A .
 - (a) Using the modified Gram-Schmidt process, write out expressions for q_1, q_2, q_3 , the first three columns of Q in the QR decomposition of A .
 - (b) Show the vector q_3 found in part (a) is orthogonal to both q_1 and q_2 .
4. Let $A = U\Sigma V^*$ be the singular value decomposition of $A \in C^{m \times n}$. Let u_j denote column j of U .
 - (a) Suppose $\text{rank}(A) = p < n < m$. Show $\{u_1, u_2, \dots, u_p\}$ is a basis for $\text{Col}(A)$ and $\{u_{p+1}, u_{p+2}, \dots, u_m\}$ is a basis for $\text{Null}(A^*)$.
 - (b) Suppose A is full rank and $x \neq 0$. Let $\sigma_i, i = 1, \dots, n$ be the singular values of A . Show

$$\sigma_1 \geq \frac{\|Ax\|_2}{\|x\|_2} \geq \sigma_n > 0.$$

If you want to use the property that $\|A\|_2 = \sigma_1$, then you must prove that.

5. If $q_1, \dots, q_n \in C^m$ is an orthonormal basis for the subspace $V \subset C^m$, prove that the orthogonal projector onto V is QQ^* , where Q is the matrix whose columns are the q_j .
6. Consider the fixed point iteration method $x_{k+1} = g(x_k)$, $k = 0, 1, \dots$ for solving the nonlinear equation $f(x) = 0$. Consider choosing an iteration

function of the form

$$g(x) = x - af(x) - b(f(x))^2 - c(f(x))^3,$$

where a, b , and c are parameters to be determined. Assume f is sufficiently differentiable and the corresponding iterations $x_{k+1} = g(x_k)$ converge to a unique fixed point z . Find expressions for the parameters a, b , and c in terms of functions of z such that the iteration method is of fourth order.

7. Consider the Lagrange polynomial approximation $p(x)$ of a function $f(x) \in C^2[a, b]$ at two points $x_0, x_1 \in [a, b]$. Determine $\alpha(x)$ explicitly in the following formula

$$f(x) - p(x) = \frac{f^{(n+1)}(\alpha(x)) \prod_{i=0}^n (x - x_i)}{(n+1)!}$$

for the case

$$f(x) = \frac{1}{x}, \quad x_0 = 1, \quad x_1 = 2.$$

8. Assume that $f \in C^3[a, b]$ and $x, x+h, x+2h \in [a, b]$. Determine constants c_1 and c_2 such that

$$\left| f'(x) - \frac{1}{h} \left[-\frac{3}{2}f(x) + c_1f(x+h) + c_2f(x+2h) \right] \right| \leq ch^2 \max_{a \leq x \leq b} |f'''(x)|,$$

where $c > 0$ is a constant independent of h .

9. Let $Q_m(f)$ denote the m -point Gaussian quadrature rule over the interval $[a, b]$ and with continuous weight function $\rho(x) \geq 0$, that is,

$$Q_m(f) = \sum_{i=1}^m \alpha_i f(x_i) \approx J(f) = \int_a^b \rho(x) f(x) dx.$$

Show that, if a and b are finite and f is continuous, then $Q_m(f) \rightarrow J(f)$ as $m \rightarrow \infty$.

10. Consider numerically solving the initial value problem

$$y'(t) = f(t, y), \quad 0 < t \leq t_f, \quad \text{with } y(0) = \eta.$$

Assume f is sufficiently differentiable and let h denote the step size. Prove that the method

$$y_{n+3} + y_{n+2} - y_{n+1} - y_n = h(f_{n+3} + f_{n+2} + f_{n+1} + f_n)$$

is consistent but not convergent.